

6.8 Quiz: Polynomials (no calculator)

1. [Maximum mark: 7]

Let $f(x) = x^2 - 4x - 12$.

(a) Write $f(x)$ in factored form, such that $f(x) = (x - a)(x - b)$. [2]

(b) Write down the x -intercepts of the graph of f . [1]

(c) Write down the y -intercept. [1]

(d) Write $f(x)$ in vertex form, i.e. $f(x) = (x - h)^2 + k$. [1]

(e) Find the coordinates of the vertex of the parabola. [2]

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2. [Maximum mark: 7]

Consider the function $g(x) = -(x - 3)^2 + 16$.

(a) Write down the coordinates of the vertex. [1]

(b) Write down the equation of the axis of symmetry. [1]

(c) Show that $g(x) = -x^2 + 6x + 7$. [2]

(d) Solve $g(x) = 0$ to find the x -intercepts. [3]

3. [Maximum mark: 7]

The line $y = 3x$ and the parabola $y = x^2 - 4$ are drawn on the same axes.

(a) Show that the x -coordinates of the points of intersection satisfy

$$x^2 - 3x - 4 = 0. \quad [2]$$

(b) Show that $x^2 - 3x - 4 = 0$ has two distinct real roots. [1]

(c) Factorise $x^2 - 3x - 4$. [1]

(d) Find the x -coordinates of the points of intersection. [1]

(e) Find the coordinates of both points of intersection as (x, y) pairs. [2]

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4. [Maximum mark: 7]

The equation $3x^2 - kx + 12 = 0$ has two equal roots.

- Write down the discriminant in terms of k . [1]
- Find the values of k . [2]
- For the positive value of k , find the repeated root. [2]
- State the values of k for which the equation has no real roots. [2]